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Smart ABET Accreditation Intelligence Platform

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Below is a **structured report-style document** you can share with your team. I wrote it in a technical but clear way, similar to how a team design document or proposal would look.

AI Integration for ABET Accreditation Support System

Overview

This project aims to integrate Artificial Intelligence capabilities into the existing university assessment software that supports programs preparing for **ABET accreditation**. The goal is to transform the current system from a data collection platform into an **intelligent decision-support platform** that can analyze performance, detect risks early, assist in reporting, and help accreditation teams prepare stronger evidence.

The AI system will introduce **four main features**:

1. **Course Risk Prediction Model**
2. **Agentic RAG Chatbot for Course Insights**
3. **Top-Performing Course Selection for ABET Evidence**
4. **Automated ABET Report Generation**

Each feature addresses a specific problem in the accreditation workflow, from predictive monitoring to documentation automation.

Feature 1: Course Risk Prediction Model

Objective

This feature predicts whether a course is **at risk of failing ABET expectations** during a given academic term. The model analyzes historical course performance data and identifies patterns associated with weak outcome attainment.

The system will also provide **explanations for the prediction**, enabling faculty and QA teams to understand the causes and apply corrective actions.

Defining the Prediction Problem

The machine learning model will perform a **binary classification task**:

Label	Meaning
0	Course is performing normally
1	Course is at risk of failing ABET expectations

A course may be labeled **at risk** if one or more of the following occurs:

- Student Outcome attainment below the required threshold
- Significant performance decline across semesters
- Low pass rate
- Weak assessment results

Dataset Structure

Each record in the dataset represents a **course in a specific academic term**.

Example dataset:

course	term	avg_score	so1	so2	so3	trend	label
DB	2023F	78	75	68	80	stable	0
Networks	2023F	62	55	60	58	down	1

Columns:

- **avg_score** – average student score in the course
- **so1, so2, so3** – attainment values for Student Outcomes

- **trend** – performance trend compared to previous term
- **label** – whether the course is at risk

Normalizing Student Outcome Scores

Since each course may have multiple Student Outcomes, we create a **composite SO performance index**.

Formula:

$$
SO_index = \frac{SO1 + SO2 + SO3 + ... + SOn}{n}
$$

Example:

Networks course:

$$
SO_index = (55 + 60 + 58)/3 = 57.67
$$

This provides a **single normalized indicator** of course outcome performance.

Trend Calculation

To determine whether course performance is improving or declining, we compare the **current SO index** with the previous term.

[
Trend_value = SO_index(current) - SO_index(previous)
]

A threshold is used to classify the trend.

Example threshold:

[
T = 3
]

Trend classification:

Condition	Trend
Trend > +3	Up
Trend < -3	Down

-3 ≤ Trend ≤ Stab
+3 le

Trend Visualization

Example:

Performance Trend

```
SO Index
80 |              *
75 |            *
70 |          *
65 |      *
60 |    *
55 | *
   ----------------------------
    2021   2022   2023   2024
```

Threshold Visualization

```
        UP TREND
           ↑
           |
```

```
threshold +3|----------------
            |
            |   stable zone
            |
threshold -3|----------------
            |
            ↓
        DOWN TREND
```

Machine Learning Models

Several models can be tested for this classification task:

- Logistic Regression (baseline)
- Random Forest
- XGBoost (recommended)

XGBoost is particularly effective for **structured tabular data**, which matches the dataset structure.

Model Explainability

To understand why the model predicts a course as risky, we will use **SHAP (SHapley Additive Explanations)**.

Example explanation output:

Course: Networks
Risk Level: High

Reasons:

- Low SO2 attainment
- Decreasing performance trend
- Low pass rate

This transparency helps faculty understand the root causes.

Feature 2: Agentic RAG Chatbot for Course Insights

Objective

Provide an intelligent chatbot that can answer questions related to:

- Course performance
- Outcome attainment
- Reasons for performance decline
- ABET report content

- Historical course data

Example queries:

- “Why did Networks course decline this semester?”
- “Which courses have weak SO3 attainment?”
- “Summarize the Database course report.”

Agentic RAG Architecture

The chatbot will use **Agentic Retrieval-Augmented Generation (RAG)**.

This means the chatbot will not rely only on a language model but will retrieve information from multiple sources before generating a response.

Sources include:

- Course reports
- Assessment documents
- Database metrics
- Outcome attainment data

System Components

1. Vector Database

Course documents and reports are converted into embeddings and stored in a **vector database**.

Possible choices:

- Qdrant
- pgvector
- Weaviate

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2. Embedding Model

Documents will be transformed into embeddings using models such as:

- BGE-large
- OpenAI text embeddings
- Multilingual embedding models if Arabic documents exist

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3. Retrieval Pipeline

When a user asks a question:

1. The question is converted into an embedding
2. Similar documents are retrieved from the vector database
3. Relevant course metrics are retrieved from SQL

4. The language model generates a grounded response

Agentic Reasoning Layer

The chatbot acts as an **AI agent** capable of deciding which tools to use.

Example tools:

- retrieve_course_metrics()
- retrieve_course_reports()
- compare_course_terms()
- retrieve_risk_prediction()

The agent decides which tool to call based on the question.

Should We Use n8n?

n8n is a workflow automation tool.

Advantages

- Easy integration with APIs
- Fast prototyping
- Automation workflows

Limitations

- Limited complex reasoning capabilities
- Harder to maintain advanced AI agents

Recommendation

For this project:

Use **Python backend with LangGraph or LangChain** for the agent logic.

n8n may optionally be used for:

- workflow automation
- scheduled tasks
- notifications

But the **core chatbot intelligence should not depend on n8n alone**.

Feature 3: Top-Performing Course Selection for ABET Evidence

Objective

During ABET accreditation, departments must present **strong evidence examples** of course performance.

This feature automatically identifies **the best-performing courses** to showcase.

Course Ranking System

Each course receives a **composite score** based on several metrics.

Example scoring formula:

$$
\begin{aligned}
&\text{CourseScore} = \\
&0.4 \times \text{SO_index} + \\
&0.2 \times \text{percentile} + \\
&0.2 \times \text{trend_stability} + \\
&0.2 \times \text{evidence_completeness}
\end{aligned}
$$

Metrics used:

Metric	Description
SO index	Overall outcome attainment

Percentile	Relative performance compared to other courses
Trend stability	Performance consistency
Evidence completeness	Availability of assessment documentation

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Output Example

course	score	reason
Database	91	high attainment and stable trend
Algorithms	88	strong outcome coverage
Signals	86	complete assessment evidence

These courses will be recommended for inclusion in **ABET documentation and review presentations**.

Feature 4: Automated ABET Report Generation

Objective

Generate ABET documentation automatically using assessment data and evidence stored in the system.

This reduces the manual effort required to prepare accreditation reports.

Report Generation Process

The report generation process follows three stages.

Stage 1: Data Aggregation

The system gathers information such as:

- course information
- student outcome attainment
- assessment results
- improvement actions
- evidence documents

Stage 2: Structured Report Template

A structured template is filled with computed metrics.

Example template section:

Course: Database Systems

Student Outcome Attainment:

SO1: 75%
SO2: 68%
SO3: 80%

Weak Outcomes:
None detected.

Improvement Actions:
Continuous monitoring recommended.

Stage 3: AI Narrative Generation

A language model improves the report by generating a **formal narrative suitable for ABET documentation**.

Example output:

The Database Systems course demonstrates strong attainment across all assessed Student Outcomes. Performance levels exceed the program threshold, indicating effective alignment between learning objectives and assessment strategies.

System Architecture Overview

The complete system architecture consists of five layers:

1. **Existing University System**
2. **Data Integration Layer**
3. **AI Analytics Layer**
4. **RAG Knowledge Layer**
5. **Dashboard and Chat Interface**

Technology Stack

Suggested tools and technologies:

Backend

FastAPI

Database
PostgreSQL

Vector Database
Qdrant or pgvector

Machine Learning
Python
Scikit-learn
XGBoost
SHAP

LLM Orchestration
LangGraph or LangChain

Visualization
Plotly
Matplotlib
Dashboard frameworks

Expected Benefits

The system provides several improvements to the accreditation workflow:

- Early detection of weak courses

- Data-driven continuous improvement
- Faster preparation of ABET documentation
- Intelligent search across course reports
- Reduced manual effort for faculty and QA teams

If you want, I can also generate a **clean visual system architecture diagram** and **trend graphs you can paste directly into the report** so it looks like a professional research/technical document.